



AQ9.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

Limits:

Level 1:

Find the value of the following limit:

$$\lim_{(x,y) \rightarrow (2,3)} \frac{x^2 + y^2 - y}{y^2 - x^2} \lim_{(x,y) \rightarrow (2,3)} y^2 - x^2 + y^2 - y$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Suppose f and g are multivariable functions defined on domain R^2 with the following limits:

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = l \text{ and } \lim_{(x,y) \rightarrow (0,0)} g(x,y) = k$$

$$f(x,y) = l \text{ and } \lim_{(x,y) \rightarrow (0,0)} g(x,y) = k$$

where $l, k \in \mathbb{R}$. Which of the following options is(are) correct?



$$\lim_{(x,y) \rightarrow (0,0)} (f(x,y) + g(x,y)) = l + k$$



$$\lim_{(x,y) \rightarrow (0,0)} (f(x,y) \cdot g(x,y)) = lk$$



$$\lim_{(x,y) \rightarrow (0,0)} (5f(x,y)) = 5l$$



$$\lim_{(x,y) \rightarrow (0,0)} \left(\frac{f(x,y)}{g(x,y)} \right) = \frac{l}{k}, \text{ when } k \neq 0$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\lim_{(x,y) \rightarrow (0,0)} (f(x,y) + g(x,y)) = l + k$$

$$\lim_{(x,y) \rightarrow (0,0)} (f(x,y) \cdot g(x,y)) = lk$$

$$\lim_{(x,y) \rightarrow (0,0)} \left(\frac{f(x,y)}{g(x,y)} \right) = \frac{l}{k}, \text{ when } k \neq 0$$

Consider the function $f: R^2 \rightarrow R$ defined as: $f(x,y) = \begin{cases} \frac{4x^2y^2}{x^2+y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$

What is the value of $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$?

(f(x,y))?

No, the answer is incorrect.

Score: 0



Accepted Answers:

(Type: Numeric) 0

1 point

Suppose f and g are multivariable functions defined on domain $\mathbb{R}^2$ as

$$f(x, y) = \frac{x^2 + y^2 + y(y - 2x - 1)}{x^2 - y^2} \quad f(x, y) = x^2 - y^2 \quad x^2 + y^2 + y(y - 2x - 1) \quad \text{and} \quad g(x, y) = \frac{2y^2 + x(x - 2y - 1)}{x^2 - y^2}$$

 $g(x, y) = x^2 - y^2 \quad x^2 + y^2 + x(x - 2y - 1)$. Find the value of $\lim_{(x, y) \rightarrow (1, 1)} (f(x, y) - g(x, y))$
 $\lim_{(x, y) \rightarrow (1, 1)} (f(x, y) - g(x, y))$.
No, the answer is incorrect.**Score: 0****Accepted Answers:**

(Type: Numeric) 0.5

1 point

Find the value of the following limit:

$$\lim_{(x, y) \rightarrow (3, 1)} \frac{x^2 - 3xy^2}{x^2 - 9y^4} \quad \lim_{(x, y) \rightarrow (3, 1)} x^2 - 9y^4 \quad x^2 - 3xy^2$$

No, the answer is incorrect.**Score: 0****Accepted Answers:**

(Type: Numeric) 0.5

1 point

Level 2:

Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as:

$$f(x, y) = \begin{cases} \frac{3xy}{x^2 + y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases} \quad f(x, y) = \begin{cases} x^2 + y^2 & 3xy \neq 0 \\ 0 & x = y = 0 \end{cases}$$

Use the given information to answer the questions 6-8:

If f approaches to L as (x, y) approaches to the origin along Y-axis, then find the value of L ?**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

If f approaches to L as (x, y) approaches to the origin along a straight line $y = 3x$, then find the value of L

No, the answer is incorrect.**Score: 0****Accepted Answers:**

(Type: Numeric) 0.9

1 point

1 point

Which of the following options is(are) false?

☐

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 \quad \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$$



$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1 \quad \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$$

☐ Limit of the given function at (0,0) exists but is not equal to 0.

☐ Limit of the given function at (0,0) does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 \quad \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1 \quad \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$$

Limit of the given function at (0,0) exists but is not equal to 0.

1 point

Which of the following functions have limit value equal to 5 at the point (1,1,-1)?



$$f(x,y,z) = \frac{x+3yz^2-zx}{xy+yz+xz^2} \quad f(x,y,z) = xy+yz+xz^2x+3yz^2-zx$$



$$f(x,y,z) = \frac{3(x+y-z)+7x^2y+yz}{xyz+zy+xz^3(x+y-z)+7x^2y+yz} \quad f(x,y,z) = xyz+zy+xz^3(x+y-z)+7x^2y+yz$$



$$f(x,y,z) = x^3 + y^3 + z^3 - 2xy - 2yz - 4zx \quad f(x,y,z) = x^3 + y^3 + z^3 - 2xy - 2yz - 4zx$$



$$f(x,y,z) = \frac{x^2e^{\frac{x}{y}} + y^2e^{\frac{y}{z}} + z^2e^{\frac{z}{x}}}{e^{\frac{x+y+z}{2}}} \quad f(x,y,z) = e^{2x+y+z}x^2e^{2x+y}y^2e^{2y+z}z^2e^{2-z}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x,y,z) = \frac{x+3yz^2-zx}{xy+yz+xz^2} \quad f(x,y,z) = xy+yz+xz^2x+3yz^2-zx$$

$$f(x,y,z) = x^3 + y^3 + z^3 - 2xy - 2yz - 4zx \quad f(x,y,z) = x^3 + y^3 + z^3 - 2xy - 2yz - 4zx$$

1 point

Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as:

$$f(x,y) = \begin{cases} \frac{x^k y}{x^{2n} + y^{2n}} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases} \quad f(x,y) = \begin{cases} x^{2n+y} y^{2n+k} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$$

where $k, n \in \mathbb{N} \setminus \{0\}$. Which of the following statements is(are) true about f ?



If $k = 2n - 1$, then the limit at (0,0) exists and is equal to 0.



If $k < 2n - 1$, then the limit at (0,0) does not exist.



If $k > 2n$, then the limit at (0,0) always exists and is equal to 0.



If $k > 2n$, then the limit at (0,0) does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $k < 2n - 1$, then the limit at (0,0) does not exist.

If $k > 2n$, then the limit at (0,0) always exists and is equal to 0.

[Check Answers](#)

Your score is: 0/10